



Respiratory responses during affective picture viewing

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Abstract

Previous research has demonstrated covariation of physiological responding with judgments of valence and arousal. However, until now links between these affective dimensions and respiratory measures have not been extensively investigated. In this study, eight picture series of different affective valence and arousal level were shown to 30 subjects, while respiration, skin conductance level (SCL), heart rate (HR) and affective judgments were measured. With increasing pleasantness, inspiratory time lengthened, mean inspiratory flow decreased and thoracic breathing increased. With increasing arousal, inspiratory time and total breath duration shortened and mean inspiratory flow, minute ventilation, thoracic breathing and electrodermal activity increased. These findings confirm the importance of arousal in respiratory responding, but also indicate a modulatory role of affective valence. We propose that the arousal effects reflect energy mobilization in preparation to act, and that the valence effects might be a manifestation of an attention bias toward negative stimuli.

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1. Introduction

Empirical work (Lang et al., 1990; Mehrabian, 1970; Osgood et al., 1957; Russell, 1980) has repeatedly confirmed that differences in affective meaning among stimuli—words,

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objects, events—can succinctly be described by three basic emotional dimensions, that is a dimension of affective valence, also called pleasure or pleasantness, ranging from positive (pleasant) to negative (unpleasant), one of arousal, ranging from calm (low-arousal) to excited (high-arousal), and one called dominance or control, ranging from controlled to in control. The valence and arousal dimensions are primary, and they typically account for most of the variance in emotional judgments (Bradley and Lang, 1994; Hamm and Vaitl, 1993). Recent work has provided evidence for a correspondance between physiological variables and the emotional dimensions of valence and arousal (e.g., Greenwald et al., 1989; Lang et al., 1993; Simons et al., 1999). However, very few studies have investigated the relation between respiratory measures and these emotional dimensions (Nykliček et al., 1997; Van Diest et al., 2001), although most available data suggest that respiration patterns may reflect general dimensions of emotional response (for review, see Boiten et al., 1994).

As to the relation between respiration and the dimensions of valence and arousal, a number of studies suggest a central role of arousal. An increase in subjective arousal appears to be accompanied by breathing acceleration and an increase in mean inspiratory flow and minute ventilation (e.g., Boiten, 1998; Boiten et al., 1994; Nakamura, 1984; Nykliček et al., 1997; Van Diest et al., 2001). On the contrary, no study has yet provided clear evidence for a relationship between inspiratory volume and valence and arousal (Boiten, 1998; Boiten et al., 1994; Van Diest et al., 2001). Pleasantness has been suggested to influence the relative contribution of the rib cage and the abdomen to inspiratory volume (Boiten et al., 1994). Yet, Boiten (1998) failed to confirm a covariation between pleasantness and thoracic-abdominal balance.

This study pursued two goals. One goal was to assess physiological reactions as they vary when looking at pictures, as defined by a priori groupings on the valence and arousal dimensions. A second goal was to assess the covariation between individual subjects' reports of valence and arousal and their physiological responses. A similar two-step analysis has been recently used by Bradley and Lang (2000).

Based on the above-mentioned evidence, we predicted respiratory variables to be mainly modulated along the arousal dimension. With increasing arousal, time parameters (inspiratory and expiratory time, total breath duration) were expected to decrease and flow parameters (mean inspiratory flow, minute ventilation) to increase. For inspiratory volume and the thoracic-abdominal ratio, no specific predictions were advanced.

Besides respiration, two other measures known to show covariation with valence and arousal were measured; skin conductance level (SCL) that we expected to increase with arousal (Greenwald et al., 1989; Lang et al., 1993) and mean heart rate (HR) that we predicted to be lower for unpleasant pictures compared to pleasant pictures (Greenwald et al., 1989; Lang et al., 1993; Palomba et al., 1997).

2. Methods

2.1. Subjects

Participants in this study were 16 men and 14 women, aged 20–28 years (mean age 23.4 years). Most participants were undergraduate students. Entry criteria for the study included good general health and no consumption of drugs that may influence the emotional

behavior. Subjects were asked to avoid drinking coffee and alcoholic beverages prior to the experiment. One further participant was excluded from analysis due to failure of the slide projector. Measures of inspiratory volume, mean inspiratory flow, and minute ventilation for two participants were not included in the analysis because the acceptance criterion for volume calibration was not fulfilled (see below). For one subject, percent rib cage could not be analyzed due to a technical failure. Thus, final sample sizes were: affective ratings, respiratory time variables, skin conductance level, and heart rate, $n = 30$; percent rib cage, $n = 29$; inspiratory volume, mean inspiratory flow, and minute ventilation, $n = 28$.

2.2. Stimuli

Forty colored pictures were chosen from the International Affective Picture System (IAPS) (CSEA, 1999; Lang et al., 1999). The pictures selected varied widely in content and affective tone, from pleasant to unpleasant and calm to arousing. The materials were presented as slide images and were arranged in eight series of five pictures each that were similar in their normative distribution of ratings.² The intent was to get eight series covering well the entire affective space. Each slide was shown for 6 s so that each series lasted 30 s. This presentation procedure was chosen for the following reasons. First, we wanted a presentation epoch long enough to include at least six to seven respiratory cycles. Second, the use of five pictures presented for 6 s each instead of one picture presented for 30 s was preferred because the normative affective ratings refer to a 6-s exposition and affective and physiological effects of 30-s stimulation with single pictures are not documented. Further, we thought that exposure to single pictures, because of their static nature, could be accompanied by a quick decrease in affective discrimination. On the contrary, by presenting five pictures of the same affective tone but of different content, differences in physiological reactivity are more likely to be maintained or even show sensitization, as Bradley et al. (1996) have reported.

The order of the five pictures in each series was randomly determined and was maintained constant throughout the entire study. Across subjects, the presentation order of the eight series was varied in such a way that each series was presented equally often as first, second, third, etc. The projected slide images were approximately 60 cm wide and 60 cm high.

² The following pictures from the IAPS were used (picture's set in parentheses). The order for each series corresponds to the order of presentation. Means $\pm$ S.D.s of the normative ratings of the five pictures for valence (V) and arousal (A) (Lang et al., 1999) are also reported. Series 1 (V: 7.45 ± 0.19 , A: 6.69 ± 0.38): erotic female 4220(2)/male 4460(3), erotic couple 4660(3), skier 8030(2), water skier 8200(3), sailing 8080(2); series 2 (V: 7.80 ± 0.25 , A: 5.03 ± 0.13): athletes 8540(5), sunset 5830(2), family 2340(3), baby 2150(2), father 2160(3); series 3 (V: 7.48 ± 0.48 , A: 3.22 ± 0.19): nature 5760(1), flowers 5200(3), rabbit 1610(1), cow 1670(1), springbok 1620(1); series 4 (V: 4.90 ± 0.23 , A: 6.14 ± 0.51): aircraft 6900(2), female kiss 4770(3), volcano 5920(3), leopard 1310(3), rock climber 8160(3); series 5 (V: 4.87 ± 0.21 , A: 2.72 ± 0.30): mushroom 5510(1), hairdryer 7050(2), neutral face 2200(1), rolling pin 7000(2), umbrella 7150(2); series 6 (V: 2.31 ± 0.22 , A: 6.69 ± 0.45): ugly wound 3030(1), plane crash 9050(2), suicide 6570(6), aimed gun 6230(2), war victim 9250(3); series 7 (V: 2.44 ± 0.37 , A: 5.35 ± 0.19): hospital 3220(3), dead seal 9180(2), dead cow 9140(2), dying man 3230(3), eye disease 3160(3); series 8 (V: 3.33 ± 0.65 , A: 3.95 ± 0.08): exhaust 9090(2), woman 9190(2), cemetery 9000(2), bag man 9331(10), puddle 9110(2).

2.3. Measurements

Judgments of affective valence, arousal, and dominance were registered using the pencil-and-paper version of the nine-point self-assessment manikin (SAM; Bradley and Lang, 1994). A possible problem associated with the presentation procedure used is that the five pictures within a series may differ in their actual affective tone, despite having very similar normative ratings. For this reason after each series subjects were given the possibility to indicate if single pictures were experienced as particularly pleasant, unpleasant, arousing, or calming.

Respiration was measured using a respiratory inductive plethysmograph (Respirace PLUS, SensorMedics, NIMS, USA). The coils of the Respirace were placed on the chest 2 cm below the nipple line (for women this was right below the breast) and on the abdomen at the umbilical level, providing the thoracic motion (V_{th}) and the abdominal motion (V_{ab}) signals. The proportionality constant between V_{th} and V_{ab} amplifiers was determined by the qualitative diagnostic calibration method (QDC) carried out during a 5-min period of spontaneous breathing (Sackner et al., 1989). The Respirace signals (in volts) were recorded with a Labview program (National Instruments, USA) and analyzed by a MATLAB program (MathWorks Inc., USA). Volume calibration of both channels was performed before and after the experiment (pre- and post-calibration) with a calibrated pneumotachograph (CPX/D, Medical Graphics Corporation, USA), employing a procedure in which the subject breathed through a mouthpiece at different rates and depths (natural, relatively fast and shallow, relatively slow and deep, natural) for about 20 s for each phase. A multiple linear regression was performed over the calibration periods, yielding the regression coefficient to be used for the reconstruction of the volume out of the Respirace signals. Calibration was accepted when the overall regression factor r^2 was over 0.8. The calculated regression coefficient was implemented into the MATLAB program, which analyzed the breathing curves and arranged the data in a log-file. Both the Respirace and the pneumotachograph signals were sampled at 200 Hz. For each breath, the following respiratory parameters were extracted: inspiratory time (T_i) and expiratory time (T_e , including expiratory pause), inspiratory volume (V_i) and expiratory volume (V_e), and the percentage of rib cage contribution to V_i (%RC). Total breath duration (T_{tot}), inspiratory duty cycle (T_i/T_{tot}), mean inspiratory flow (V_i/T_i) and minute ventilation (MV, calculated by $V_i/T_{tot} \times 60$ s/min) were calculated off-line. Only V_i was used as a measure of tidal volume. In comparison to the traditional analysis of respiratory rate and depth, this more detailed breathing pattern analysis provides a better insight in the mechanisms that regulate respiration. In fact, there is evidence for a respiratory control model that incorporates an inspiratory drive mechanism and a phase-switching mechanism, which initiates as well as terminates inspiratory activity. Mean inspiratory flow and inspiratory duty cycle are considered to be indices of the activity of these mechanisms (for review, see Wientjes et al., 1998).

Skin conductance level and heart rate were measured with Varioport (Becker Meditec, Karlsruhe, Germany), an eight-channel recording system. After amplification and filtering, data were digitized (12 bit resolution) and saved on a compactflash-card. The device was controlled via computer running VP-Basic software (TEMEC Instruments B.V., 6460 HA Kerkrade, NL) and the channels were defined as follows: SCL (unit μ S): sampling rate, 256 Hz; saving rate, 16 Hz; range, 0.1–100 μ S; resolution, 0.001 μ S; ECG (unit mV): sam-

pling rate, 512 Hz; saving rate, 256 Hz; resolution, 1 μ V; pulse (unit beats/min): sampling rate, 512 Hz; saving rate, 4 Hz; resolution, 1 beat/min. Pulse was determined on-line from the ECG signal. SCL was recorded using a 16 bit resolution unit, which provided a constant 0.5 V across electrodes. Before recording, the palm of the non-dominant hand was cleansed with distilled water and two Ag/AgCl miniature electrodes (4 mm contact area diameter, GE Medical Systems, Milwaukee, USA) filled with lubricating jelly (MED-TEK/Synapse, Arcadia, CA 91007, USA) were placed on the thenar and hypothenar eminence of the palm. For ECG recordings, pregelled Ag/AgCl electrodes (60 mm, Skintact[®], Leonherd Lang GmbH, Innsbruck, Austria) were placed on the sternum and the left costal arch at the level of the 10th rib, with a reference below the right collar-bone.

2.4. Procedure

Participants were tested individually in one experimental session lasting approximately 1 h. The experiment took place in a noiseless, air-conditioned room. After arrival, participants filled out an informed consent and the male experimenter provided them with an outline of the experimental procedure. Participants were told that eight slide series consisting of five pictures of the same normative distribution of ratings would be displayed, and that each picture should be attended for the entire duration of presentation. They were also told that the entire series as a whole had to be rated and not the single pictures but that they could indicate if single pictures within a series were experienced as particularly negative, positive, calming, or arousing. The importance of rating the slide series as they actually felt while they watched the pictures was emphasized. Next, they were explained how to use the SAM rating scales and two pictures were displayed in order to familiarize them with the SAM rating procedure. Participants sat in a comfortable armchair, at approximately 1.2 m from the projection wall, leaning on the back of the seat, with both feet on the floor and the arms lying on a table.

Following attachment of bands and electrodes, the QDC-calibration and the pre-calibration with the pneumotachograph were performed. Afterwards, the lights were dimmed and a 2-min resting baseline was initiated before the first slide series. Five seconds before each slide series, participants were verbally informed by the experimenter that the next series would be shown. After offset of the fifth picture, participants immediately rated the series and then relaxed for approximately one minute till the next series was announced. After the last rating, lights were switched on again and post-calibration was carried out. Bands and sensors were removed, and participants were thanked and paid for their participation. Before leaving, they were asked not to divulge any of the details of the experiment to other potential participants.

2.5. Data reduction

A preliminary analysis was conducted analyzing the entire presentation epoch and the first and the second 15 s separately. The major effects in terms of affective discrimination in respiratory reactivity were obtained for the second half of series presentation. This suggests that affective discrimination increased across time (Bradley et al., 1996). In the following, we report the results for the second half of series presentation. For the breathing parameters,

median values of the 25-s interval before stimulus announcement (baseline) and median values of the 15-s interval before stimulus offset (stimulus interval) were computed for each subject and each slide series. These intervals were calculated from the end of the last full respiratory cycle before stimulus announcement and before stimulus offset, respectively. The beginning of the intervals fell always within a respiratory cycle. This cycle was entirely included in the intervals so that intervals were slightly longer than 25 and 15 s, respectively. Over all subjects, baselines had a mean duration of 26.3 ± 0.7 s and included on average 6.6 ± 1.2 respiratory cycles, whereas stimulus intervals had a mean duration of 17.1 ± 0.9 s and included on average 4.6 ± 1.1 respiratory cycles. Change scores were computed by subtracting the baseline median from the median of the stimulus interval. Similarly, SCL and HR were computed by subtracting the mean value of the 25-s interval before stimulus announcement from the mean value of the last 15 s of series presentation. For the respiratory parameters, we preferred to use the median instead of the mean because the median is a much more robust estimate of the center of a sample of data.

2.6. Statistical analysis

Two analyses were conducted, one for each goal given in the introduction. The eight slide series were selected to constitute two independent variables with three levels each: “valence” (positive, neutral, negative) and “arousal” (low, moderate, high). The emotional stimuli were selected well in respect to a representative norm (Lang et al., 1999). According to the normative ratings, slide series 1–3 included positively valent pictures; slide series 4 and 5 neutrally valent pictures; slide series 6–8 negatively valent pictures. Series 1, 4, and 6 were of high arousal; series 2 and 7 of moderate arousal; series 3, 5, and 8 of low arousal. Therefore, one would expect the relationship between physiological responses and valence and arousal to be available from comparisons of specific slide series. This was done in analysis 1.

On the other hand, people can differ in their affective reaction to stimuli. Although there is often an affective response that is typical for the majority of individuals, there are always individuals whose emotional response differs from that of the majority. It is important to take into account this individuality in affective reactions to stimuli. This was done in analysis 2 by using an analytic strategy that correlates directly physiological responses with the actual ratings of valence and arousal obtained from the participants.

2.6.1. Analysis 1: Reactions to *a priori* valence and arousal categories

Polynomial contrasts of first order (linear trends) were tested along four combinations of three-slide series for each physiological variable. Based on the normative classification, the three slide series of each combination were similar in one of the variables valence and arousal, but differed in the other one. The three slide series were ordered along the variable in which they differed (for valence (V): positive—neutral—negative, for arousal (A): high—moderate—low). The four combinations of three-slide series were:

1. Slide series 1 (positive V, high A), 2 (positive V, moderate A), and 3 (positive V, low A). These slide series were labeled “positive series”.

2. Slide series 6 (negative V, high A), 7 (negative V, moderate A), and 8 (negative V, low A). These slide series were labeled “negative series”.
3. Slide series 1 (positive V, high A), 4 (neutral V, high A), and 6 (negative V, high A). These slide series were labeled “high-arousal series”.
4. Slide series 3 (positive V, low A), 5 (neutral V, low A), and 8 (negative V, low A). These slide series were labeled “low-arousal series”.

The first and second contrasts were expected to reveal linear relationships with arousal for positive and negative pictures, respectively. The third and fourth contrasts should indicate if there were linear relationships with valence for high- and low-arousal pictures, respectively.

The same four polynomial contrasts were also tested on the valence and arousal ratings to determine if mean ratings of slide series were in accordance with expectations. Statistical significance was evaluated at the 0.05 level.

2.6.2. *Analysis 2: Covariation of affective ratings and physiological responses*

The second analysis assessed the covariation between subjects' ratings of valence and arousal and their physiological responses. For this analysis, the median values of baselines and stimulus intervals were first log-transformed to minimize skewness and heteroscedasticity, in accordance with standard statistical practice and consistent with previous studies (e.g., Fridlund, 1991; Jäncke, 1994). Change scores were calculated by subtracting the transformed baseline score from the transformed score of the stimulus interval. Change scores were further multiplied by 1000.

The “summary statistics approach” (Davis, 2002), which has often been used under other names too, was used for this analysis. We proceeded as follows. The model $Y = \alpha + \beta_V V + \beta_A A + E$ was fitted for each subject separately. Y is the response score of the physiological measures as defined above. V and A are the subjective valence and arousal judgments given by the participants and E is the random error parameter. The model was fitted using an M -estimator with a psi function of the Huber type and the default tuning constant of 1.345 (Venables and Ripley, 2002). The coefficients (slope) β_V and β_A were estimated for each subject separately, and these numbers were tested by two Wilcoxon signed rank tests against the null hypothesis of symmetry around zero. Statistical significance was evaluated at the 0.05 level. The median of the subjects' coefficients was used as estimation of the effect. A positive coefficient indicates a positive linear relation, and a negative coefficient indicates a negative linear relation.

This statistical analysis was also used to test the linear correlation between ratings of valence and ratings of arousal.

The summary statistics approach was preferred to the more traditional multilevel regression analysis because examination of the estimated coefficients for the individual subjects showed that their distribution was often not normal, thus violating a basic assumption of multilevel regression analysis. This fact also explains why we preferred the Wilcoxon signed rank test to the commonly used t -test. Alternatively, we could have accommodated the data by removing outlying observations, with the disadvantage, however, of losing data. A main feature of the “summary statistics approach” is its flexibility, which allows to introduce robust methods, as described.

3. Results

3.1. Analysis of reactions to *a priori* valence and arousal categories

3.1.1. Affective judgments

3.1.1.1. Intra-stimulus homogeneity. The intra-stimulus homogeneity with respect to the perceived levels of valence and arousal was good. Subjects reported one picture within a series to be particularly pleasant, unpleasant, arousing, or calming 146 times. Only six of these indications (4%) were not completely concordant with the ratings for the entire series (e.g., picture 4 of series 7 was indicated by one subject as particularly arousing with an arousal rating for the entire series of 4).

3.1.1.2. Mean ratings. Means and S.E.M.s of the valence and arousal ratings for the eight slide series are presented in Table 1. For the positive series, arousal ratings decreased from series 1 to 3 ($F(1, 29) = 132.49$, $P < 0.000001$), and valence ratings increased slightly ($F(1, 29) = 6.37$, $P < 0.05$). For the negative series, arousal ratings decreased slightly from series 6 to 8 ($F(1, 29) = 6.72$, $P < 0.05$), and valence ratings increased ($F(1, 29) = 18.79$, $P < 0.001$). For the high-arousal series, valence ratings decreased from series 1 to 4 to 6 ($F(1, 29) = 253.77$, $P < 0.000001$), whereas for arousal ratings the linear trend was not significant ($F(1, 29) = 1.79$). Finally, for the low-arousal series, valence ratings decreased in a monotonic fashion from series 3 to 5 to 8 ($F(1, 29) = 202.90$, $P < 0.000001$), whereas arousal ratings increased ($F(1, 29) = 36.95$, $P < 0.00001$).

3.1.1.3. Correlation between subjective ratings of valence and arousal. Subjective ratings of valence and arousal were linearly correlated (median coefficient of correlation was -0.37 , $P < 0.001$, Wilcoxon signed rank test). Thus, increases in unpleasantness were associated with higher arousal.

3.1.2. Physiological reactions

Table 1 presents the mean change scores in all physiological measures for the eight slide series. Significant linear trends were obtained for the following measures. T_i and T_{tot} lengthened across the positive series 1–3 in a monotonic fashion (T_i : $F(1, 29) = 7.19$, $P < 0.05$; T_{tot} : $F(1, 29) = 6.95$, $P < 0.05$). V_i/T_i decreased from series 1 to 2 to 3 ($F(1, 29) = 16.31$, $P < 0.001$) and increased across the low-arousal series 3, 5, and 8 ($F(1, 29) = 10.83$, $P < 0.01$). MV decreased from series 1 to 2 to 3 ($F(1, 29) = 16.33$, $P < 0.001$) and across the negative series 6–8 ($F(1, 29) = 4.72$, $P < 0.05$), whereas it increased from series 3 to 5 to 8 ($F(1, 29) = 6.13$, $P < 0.05$). SCL decreased from series 1 to 2 to 3 ($F(1, 29) = 15.08$, $P < 0.001$), from series 6 to 7 to 8 ($F(1, 29) = 6.61$, $P < 0.05$) and across the high-arousal series 1, 4, and 6 ($F(1, 29) = 5.54$, $P < 0.05$). For all other measures (T_e , T_i/T_{tot} , V_i , %RC, and HR), no significant linear contrasts across any of the four three-slide series combinations were obtained.

3.2. Covariation of affective ratings and physiological responses

Table 2 presents the estimated regression coefficients β_V and β_A and the interquartile range as a measure of their variability for all physiological measures. Significant effects

Table 1
Means and S.E.M.s (in parentheses) of affective ratings and changes in physiological measures for the eight slide series

	Slide series							
	1	2	3	4	5	6	7	8
Valence	7.30 (0.25)	7.87 (0.24)	7.97 (0.21)	5.33 (0.26)	5.30 (0.20)	2.00 (0.19)	1.97 (0.18)	3.13 (0.23)
Arousal	7.07 (0.25)	4.03 (0.38)	2.43 (0.32)	6.23 (0.24)	2.77 (0.31)	6.57 (0.36)	5.87 (0.32)	5.30 (0.35)
T_i (s)	−0.37 (0.17)	−0.07 (0.07)	0.16 (0.13)	−0.09 (0.09)	−0.15 (0.09)	−0.22 (0.09)	−0.22 (0.10)	−0.14 (0.09)
T_e (s)	−0.26 (0.11)	−0.20 (0.10)	−0.06 (0.11)	−0.30 (0.09)	−0.11 (0.12)	−0.02 (0.13)	−0.25 (0.08)	−0.13 (0.11)
T_{tot} (s)	−0.63 (0.19)	−0.29 (0.13)	−0.00 (0.16)	−0.51 (0.17)	−0.28 (0.18)	−0.19 (0.16)	−0.42 (0.13)	−0.26 (0.13)
T_i/T_{tot} (−)	−0.020 (0.016)	0.015 (0.011)	0.013 (0.012)	0.017 (0.011)	−0.017 (0.014)	−0.032 (0.016)	−0.012 (0.013)	−0.001 (0.015)
V_i (ml)	8 (26)	1 (21)	−9 (18)	27 (26)	−29 (27)	25 (26)	7 (23)	−6 (17)
V_i/T_i (ml/s)	56 (16)	9 (15)	−38 (13)	29 (19)	4 (11)	52 (18)	55 (15)	14 (16)
MV (l/min)	1.41 (0.33)	0.43 (0.35)	−0.25 (0.26)	1.27 (0.42)	0.24 (0.28)	1.34 (0.33)	1.31 (0.27)	0.54 (0.33)
%RC (%)	−2.2 (1.0)	−2.6 (0.8)	−2.6 (1.0)	−2.3 (1.2)	−2.9 (0.9)	−2.5 (1.0)	−2.6 (0.8)	−3.3 (0.7)
SCL (μ S)	0.201 (0.058)	−0.036 (0.033)	−0.002 (0.027)	0.075 (0.043)	−0.064 (0.026)	0.099 (0.059)	−0.020 (0.030)	−0.049 (0.025)
HR (bpm)	−0.82 (0.70)	0.34 (0.70)	0.83 (1.00)	0.42 (0.55)	−0.62 (0.73)	−1.46 (0.71)	−0.12 (0.52)	−0.76 (0.57)

Notes: For valence and arousal, higher values indicate more positive and more aroused, respectively, on a 1–9 scale. T_i : inspiratory time; T_e : expiratory time; T_{tot} : total breath duration; T_i/T_{tot} : inspiratory duty cycle; V_i : inspiratory volume; V_i/T_i : mean inspiratory flow; MV: minute ventilation; %RC: percent rib cage; SCL: skin conductance level; HR: heart rate. According to the normative ratings (see footnote 1), slide series 1–3 included positively valent pictures; slide series 4 and 5 neutrally valent pictures; slide series 6–8 negatively valent pictures. Series 1, 4, and 6 were of high arousal; series 2 and 7 of moderate arousal; series 3, 5, and 8 of low arousal.

Table 2

Medians of the estimated regression coefficients β_V (valence) and β_A (arousal) for all physiological measures obtained with analysis 2

Dependent variable	Valence estimate coefficient	Arousal estimate coefficient
T_i	+9.5* (15.5)	−3.8* (29.0)
T_e	−2.9 (25.0)	−2.8 (19.3)
T_{tot}	−0.4 (21.1)	−3.7* (21.0)
T_i/T_{tot}	+2.4 (12.8)	−0.1 (16.2)
V_i	+0.1 (14.7)	+6.8 (14.6)
V_i/T_i	−15.9** (18.2)	+13.4*** (24.5)
MV	−4.2 (19.0)	+12.2** (26.8)
%RC	+2.6* (4.6)	+3.3* (10.1)
SCL	+0.6 (2.7)	+1.9** (5.7)
HR	+1.4 (4.3)	−0.2 (5.7)

Notes: T_i : inspiratory time; T_e : expiratory time; T_{tot} : total breath duration; T_i/T_{tot} : inspiratory duty cycle; V_i : inspiratory volume; V_i/T_i : mean inspiratory flow; MV: minute ventilation; %RC: percent rib cage; SCL: skin conductance level; HR: heart rate. (*) $P < 0.05$; (**) $P < 0.01$; (***) $P < 0.001$ (Wilcoxon signed rank test against the null hypothesis of symmetry around zero). Interquartile ranges are in parentheses.

were obtained for T_i , T_{tot} , V_i/T_i , MV, %RC, and SCL. With increasing arousal, T_i and T_{tot} shortened. T_i also correlated with rated valence, in such a way that T_i lengthened with increasing pleasantness. V_i/T_i increased with arousal and decreased with pleasantness. MV was higher during high-arousal than during low-arousal slide series, but the valence effect only approached the significance level. %RC was found to increase both with ratings of valence and of arousal. Increases in rated arousal were accompanied by increases in SCL. For T_e , T_i/T_{tot} , V_i , and HR no significant correlation either with valence ratings or with arousal ratings was obtained.

4. Discussion

The aim of this study was to assess the relationships between respiratory measures and the affective dimensions of valence and arousal. Two analyses were conducted, one based on a priori defined valence and arousal categories and the second one correlating physiological responses with self-reported ratings. As to the first analysis, the intended manipulation of valence and arousal was not completely successful. The mean arousal rating of series 8 was much higher than expected from the normative ratings, making impossible to unambiguously attribute the effects obtained for the two contrasts including series 8 to either valence or arousal. Therefore, only the results from the two other contrasts could be considered.

Fig. 1 gives a summary of the significant relationships obtained in the second analysis. With increases in arousal, T_i and T_{tot} shortened and V_i/T_i , MV and SCL increased. These relationships were also found in the first analysis for the positive slide series. These findings confirm and extend previous reports on respiratory rate and flows (Boiten, 1998; Boiten et al., 1994; Nakamura, 1984; Nyklíček et al., 1997; Van Diest et al., 2001) and electrodermal activity (Greenwald et al., 1989; Hamm and Vaitl, 1993; Lang et al., 1993). Viewing of high-arousal pictures was further accompanied by enhanced thoracic breathing.

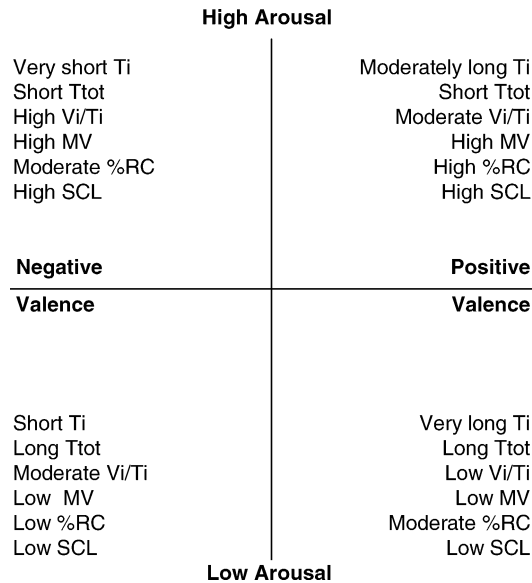


Fig. 1. Physiological response patterns for the four quadrants of the two-dimensional affective space defined by valence (positive/negative) and arousal (high/low). T_i : inspiratory time; T_{tot} : total breath duration; V_i/T_i : mean inspiratory flow; MV: minute ventilation; %RC: percent rib cage; SCL: skin conductance level.

We also observed fast predominantly thoracic breathing when listening to exciting music (Gomez and Danuser, 2004) and Rehwooldt (1911) reported more thoracic breathing during imagery of excited positive affects compared to calm positive affects. Interestingly, abdominal/diaphragmatic slow breathing is an integral part of several meditation and relaxation techniques (Arambula et al., 2001; Lowenstein, 2002).

T_i , V_i/T_i , and %RC showed to be also influenced by the affective valence of the stimuli. With increasing pleasantness, T_i lengthened, V_i/T_i decreased, and %RC increased. In a study with environmental noises and music (Gomez and Danuser, 2004), we found quite different results. During exposition to environmental noises, no influence of pleasantness on any of the three parameters was found, whereas during exposition to music, T_i shortened and V_i/T_i increased with increases in pleasantness, and no relationship was observed for %RC. These stimuli differ primarily from nonmoving pictures in that they are dynamic, that is, information is added serially and, in particular for music, with a certain rhythm. Part of the respiratory responses to these stimuli very likely reflects differences in their dynamics and rhythms. So, for music, positive excerpts had faster tempo and more dance-like rhythms than negative excerpts. On the other hand, the information in a static picture is presented all at once and essentially remains the same during presentation. Because of that, the use of pictures is advantageous when we want to investigate respiratory changes that are not due to the dynamical features of the stimulus.

To our knowledge, no previous study has reported an influence of valence on time, volume or flow parameters of respiration (cf. Boiten, 1998). According to Boiten et al. (1994), findings from some older studies may indicate a valence dependency of %RC, however in

the opposite direction that we have found (Ancoli and Kamiya, 1979; Ancoli et al., 1980; Faulkner, 1941; Svebak et al., 1981). Besides possible methodological shortcomings (cf. Boiten, 1998), it is difficult to say if the changes in thoracic-abdominal ratio reported in these studies actually reflect changes in pleasantness, or rather in arousal level, or in both.

As noted by Lang et al. (1993, pp. 269–270), looking at pictures is an aesthetic task in which the participant engages willingly and probably less strongly activates basic appetitive and defensive motives than life's "real" transaction. Thus, one could wonder if the relationships found between the valence and arousal self-reports and the physiological measures reflect differences in the emotional impact of the pictures or rather differences in the attentional involvement. In relation to the interaction between attention and emotion, it has been proposed that attention is more deeply engaged in events that have some type of affective/motivational value than in routine, neutral events (Lang et al., 1997), so that, emotion and attention are in fact closely interrelated processes.

According to classical interpretations, physiological changes of emotions prepare the body for action (Hamm et al., 2003) and the specific response patterns depend on the action tendency (e.g., flight, fight, immobility, exploration, consumption). The increase in skin conductance with increasing arousal found in this study points to a progressive activation of the sympathetic branch of the autonomic nervous system, which prepares the organism for action by mobilizing energy. Thus, the physiological changes along the arousal dimensions likely reflect increasing readiness to act (Lang et al., 1997).

More difficult appears to us the interpretation of the valence effect observed for T_i , V_i/T_i and %RC. No difference in skin conductance level between positive and negative pictures was found, excluding that the sympathetic system was differently activated. Moreover, T_i was shorter and V_i/T_i was higher for unpleasant pictures, whereas %RC was enhanced for pleasant pictures, that is, the relation between the three parameters was not the same as found for arousal, where shorter T_i , increased V_i/T_i and enhanced %RC were all associated with high arousal. A possible explanation of the valence effects can be advanced in terms of attention allocation. Researchers have consistently found that in general negative stimuli have a greater impact than positive stimuli (for reviews see Baumeister et al., 2001; Peeters and Czapsinski, 1990; Rozin and Royzman, 2001). This heightened sensitivity to negative information is known as negativity bias and has been found in several domains of human behavior. A growing body of literature is documenting an attention bias toward negative information. That is, negative information seems to command more attention. The privileged access of negative stimulation to attentional resources has been suggested by studies on autonomic physiological responses (Öhman et al., 1993), behavioral studies (Fiske, 1980; Pratto and John, 1991), and more recently by studies on brain activity using pictorial stimuli of different pleasantness but equal arousal (Carretié et al., 2003; Carretié et al., 2001; Ito et al., 1998; Smith et al., 2003). Thus, it could be that the valence effect found for T_i , V_i/T_i and %RC reflects an attention bias toward the negative pictures. Interestingly, the activity of the amygdala is greater for negative than for positive pictures (Lane et al., 1997). Extant evidence is consistent with the argument that the amygdala is critical for recruiting and coordinating cortical arousal and vigilant attention for optimizing sensory and perceptual processing of stimuli that have major significance for the individual and particularly of those associated with undetermined contingencies (Davidson, 2003). Electrical stimulation of the amygdaloid central nucleus of animals has been shown to modify their breathing

patterns (Applegate et al., 1983; Harper et al., 1984). In particular, Harper et al. (1984) have reported a shortening of inspiratory time. As to %RC, one may argue that reduced thoracic breathing while processing the negative pictures reflects a tightening of the chest wall muscles which could well be part of an avoidance/defense response. Yet, a shift toward abdominal breathing may have a more adaptive benefit. Abdominal movements, as opposed to thoracic movements, may produce less perceptible sensations of breathing and may, therefore, not interfere as much with the attentive processing of significant information. In this regard, it is worth noting that focusing attention to the pictures was consistently accompanied by a clear decrease in %RC (see Table 1), suggesting a relationship between attentional involvement and the thoracic-abdominal balance. The same phenomenon was also observed with film clips (Gomez et al., 2004).

Contrary to expectation, we did not find heart rate to decrease with unpleasantness, although the trend was in the expected direction. According to Bradley (1998), a valence modulation is generally not observed when subjects have prior knowledge of picture affect. In our study, subjects were told that the five pictures of each series were expected to be similar in their affective tone. Thus, after the first picture of each series subjects could have developed an expectancy about the affective content of the following pictures so that affective novelty was markedly reduced in the second half of series presentation. We can not exclude that the respiratory results were affected by possible expectancy effects.

In conclusion, our study has shown that breathing responding during picture viewing is organized to a certain degree along the affective dimensions of valence and arousal. For T_i , T_{tot} , V_i/T_i , MV, and %RC, a covariation with arousal was observed, confirming the importance of this affective dimension in respiratory responding. A modulatory role of valence was found for T_i , V_i/T_i and %RC. Future studies should test the validity of our findings with longer stimulus epochs and try to determine the possible origin of these relationships. We suggested that the arousal dependency likely reflects differences in energetic activation, whereas the valence dependency might be a manifestation of an attention bias toward negative stimuli.

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